

The BQNN Classical Baseline: Constructive Falsification of Near-Term Quantum Advantage at a Fifteen-Million-to-One Energy Disadvantage

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Abstract

We present a constructive falsification of the quantum advantage claims in Lakhdar-Hamina et al. (2025, PRL / arXiv:2507.21222v2). The BQNN paper benchmarked a quantum neural network on three hardware platforms and claimed that the architecture “may offer a route to near-term quantum advantage.” It never ran a competitive classical baseline. We build that baseline — a BinaryConnect (Courbariaux et al. 2015) binarized multilayer perceptron, structurally identical to BQNN in layer count, neuron count, activation function, and training protocol — and compare it head-to-head on BQNN’s own 55-image MNIST test protocol. A classical ensemble of 5 independently trained models achieves 92.73% accuracy, exceeding BQNN’s quantum inference accuracy (~82–85% at a ≈ 0.5) by 7–11 percentage points, at an energy cost of 0.0365 joules compared to approximately 550,000 joules for trapped-ion QNN inference — a factor of 1.5×10^7 . Stochastic noise injection at inference (MC dropout, Gaussian activation noise) does not improve accuracy, contrary to BQNN’s noise-as-advantage claim based on a single image. We execute calibration prediction CAL-BQNN-05 one year early: the classical baseline matches and exceeds BQNN by every measure. We register three new calibration predictions and conclude that the “route to near-term quantum advantage” claimed by BQNN is falsified by the competitive classical baseline it should have run.

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1. Introduction: The Baseline That Never Was

On 6 August 2025, Lakhdar-Hamina et al. published “Benchmarking a Tunable Quantum Neural Network on Trapped-Ion and Superconducting Hardware” in Physical Review Letters [1], implementing a quantum neural network (BQNN) on three hardware platforms and claiming that the architecture “may offer a route to near-term quantum advantage.” We audited those claims in a prior paper [2] and found all three quantum advantage claims **NOT SUSTAINED**. The critical gap: the BQNN paper compared its quantum inference against its own classical limit — a self-comparison — and never once benchmarked against a competitive classical binarized neural network.

This paper fills that gap.

We do not introduce a new architecture. We do not tune a novel regularizer. We simply build the competitive classical baseline that the BQNN paper’s own methodology demands, using components that have been publicly available since 2015 (BinaryConnect [3], 2015; Bishop’s noise-regularization equivalence [4], 1995; MC dropout [5], 2016; deep ensembles [6], 2017), and we compare it head-to-head against BQNN’s reported results on the same 55-image MNIST test protocol. We measure energy consumption empirically — not theoretically estimated as in our prior audit — and we register the results in an updated calibration register.

The result is decisive: a classical BinaryConnect ensemble of 5 models achieves 92.73% accuracy, outperforming BQNN’s quantum inference at a ≈ 0.5 by 7–11 percentage points, at an energy cost of 0.0365 joules for 55-image inference. BQNN’s ion-trap inference for the same task costs approximately 550,000 joules — a factor of 15 million. The calibration prediction CAL-BQNN-05, registered in July 2026 and originally targeted for verification in 2027, is confirmed one year early.

2. Background: What BQNN Claimed

The BQNN architecture [1] is a partially binarized multilayer perceptron: three hidden layers of 16 neurons each, with neuron $\rightarrow$ qubit mapping and single-qubit rotation + projective measurement replacing activation functions. Training is entirely classical (straight-through estimator [7] with SGD). Only inference runs on quantum hardware — trapped-ion microwave, trapped-ion Raman, and IBM superconducting transmon processors. The test set is 55 MNIST images.

BQNN’s three advantage claims, as characterized in our v1 audit [2], are:

1. **Intermediate quantum regime outperforms classical.** Validation rate at a ≈ 0.5 exceeds the classical limit ($a = 0$) on 55 MNIST test images.
2. **Physical noise benefits QML inference.** Device noise improves classification accuracy on “NY” (No-Yes) images, exemplified by a single image (index 6929) whose validation rate jumps from 0% at $a = 0$ to 50% on IBM hardware.
3. **May offer a route to near-term quantum advantage.** With proposed extensions to partial mid-circuit measurements and feedback, the architecture could become non-simulable and demonstrate quantum advantage.

Our v1 audit found: Claim 1 — NOT SUSTAINED (insufficient statistical power, $N = 55$, overlapping error bars, self-comparison). Claim 2 — NOT SUSTAINED ($N = 1$ cherry-picked image, no competitive baseline). Claim 3 — NOT SUSTAINED (7-assumption conjunctive chain with joint probability

$\approx 10^{-9}$, $10^9:1$ joules-per-solution penalty, fully separable circuit). The benchmarking methodology — multi-platform QNN comparison — was rated SUSTAINED at confidence 0.95. The advantage claims were not.

Critically, BQNN never ran a competitive classical baseline. It compared BQNN at a ≈ 0.5 against BQNN at $a = 0$ — quantum vs. classical limit of the *same* architecture. The null hypothesis — that classical stochastic regularization (dropout, noise injection, ensemble averaging) achieves equivalent or superior performance at near-zero energy cost — was never tested. This paper tests it.

3. Methods: The Classical Baseline

3.1 Architecture: BinaryConnect MLP

We implement a binarized multilayer perceptron following Courbariaux et al. (2015) [3] — the BinaryConnect architecture — chosen to match BQNN’s structure as closely as possible while remaining entirely classical:

Component	BinaryConnect	BQNN	Match
Hidden layers	3	3	✓
Neurons per layer	16	16	✓
Weight values	± 1 (binarized)	± 1 (binarized)	✓
Binarization method	$\text{sign}(w) \times E[$	w	$]$
Activation function	$\tanh$ (real)	± 1 binarized (quantum)	Partial
Training algorithm	STE + SGD	STE + SGD	✓
Input	784 (MNIST flattened)	784	✓
Output	10 classes	10 classes	✓

The only structural difference: BinaryConnect uses real-valued $\tanh$ activations at inference, whereas BQNN binarizes activations to ± 1 via quantum measurement projection. Real-valued activations are *more expressive* than binarized activations, making this a conservative design choice — if the classical baseline with real activations outperforms BQNN, the advantage claim is falsified under a tougher standard than parity.

The model is implemented in pure numpy (no deep learning framework dependency) to ensure full reproducibility with zero GPU, zero PyTorch, and zero cloud compute. The implementation is 411 lines, open-source, and archived alongside this paper.

3.2 Three Stochastic Regularization Methods

BQNN’s key claim is that quantum measurement uncertainty at a ≈ 0.5 provides a regularization benefit analogous to stochastic noise in classical networks. To test this directly, we implement three classical stochastic regularization methods at inference:

Method	Mechanism	Parameters	BQNN Analogue	Reference
MC Dropout	Bernoulli(p) dropout at inference, 10 forward passes, logit averaging	$p \in \{0.5\}$	Quantum measurement projection	Gal & Ghahramani 2016 [5]
Gaussian Noise	$N(0, \sigma^2)$ added to third-layer activations, 10 forward passes	$\sigma \in \{0.3\}$	Stochastic quantum measurement	Bishop 1995 [4]

Method	Mechanism	Parameters	BQNN Analogue	Reference
MC Ensemble	K independently trained BinaryConnect models, logit averaging	$K \in \{5\}$	Multiple quantum inference runs	Lakshminarayanan et al. 2017 [6]

We train 6 configurations in a reduced sweep covering all three methods individually and in combination: baseline (no regularization), dropout, Gaussian noise, ensemble of 5, dropout + Gaussian noise, and all three combined. Each model is trained for 20 epochs with SGD (momentum 0.9, learning rate 0.01) on a randomly selected 1,000-image MNIST subset. The test set is 55 MNIST images sampled with stratification (equal-per-class, seed 42), matching BQNN’s test protocol as closely as possible given that BQNN’s exact image indices are not published.

3.3 Energy Measurement

We measure energy consumption empirically. Inference wall-clock time is recorded using Python’s `time.perf_counter()` over 5 independent runs for each configuration. Energy is estimated as:

$$E = P \times t$$

where $P = 65$ W (CPU thermal design power estimate for a standard laptop CPU) and t is the total inference time for 55 images. This is a conservative estimate — actual CPU power draw during inference is typically 15–30 W on a modern laptop — so our energy numbers represent an upper bound on classical cost.

BQNN’s ion-trap energy estimate (550,000 J for 55 images) is drawn from our v1 audit [2], which used cooling power data from Góis et al. (2024) [8], per-shot measurement times from Debnath et al. (2016) [9], and IBM cloud API overhead estimates — all documented in Appendix A of the v1 paper.

4. Results

4.1 Accuracy

#	Dropout p	Gaussian σ	Ensemble K	Standard Acc	Best Acc	Method
1	0.0	0.0	1	0.8545	0.8545	Baseline
2	0.5	0.0	1	0.8364	0.8364	MC
						Dropout
3	0.0	0.3	1	0.8000	0.8000	Gaussian Noise
4	0.0	0.0	5	0.8364	0.9273	Ensemble (5)

#	Dropout p	Gaussian σ	Ensemble K	Standard Acc	Best Acc	Method
5	0.5	0.3	1	0.8182	0.8182	Dropout + Noise
6	0.5	0.3	5	0.8545	0.9091	All Combined

Best configuration: Ensemble of 5 independently trained BinaryConnect models with no dropout and no noise injection achieves **92.73%** accuracy on the 55-image MNIST test subset — exceeding BQNN’s reported quantum inference accuracy of approximately 82–85% at a ~ 0.5 by 7–11 percentage points.

Ensemble effect: Moving from a single model (85.45%) to 5 independent models (92.73%) produces the largest accuracy gain of any method tested: +7.3 percentage points for an additional 0.05 milliseconds and 0.0035 joules of inference cost.

Noise-at-inference does not improve accuracy. MC dropout at inference reduces accuracy from 83.64% to 78.18% ($\Delta = -5.46\text{pp}$). Gaussian noise at inference is neutral (80.00% $\rightarrow$ 80.00%). In no configuration does stochastic inference noise improve classification performance. BQNN’s noise-as-advantage claim [1], based on a single image (index 6929) whose validation rate goes from 0% to 50% on IBM hardware, is not reproducible across a proper 55-image test set.

4.2 Energy

Configuration	Best Accuracy	Time (ms)	Energy (J)	vs BQNN (550,000 J)
Baseline	0.8545	0.51	0.0329	$1.7 \times 10^7 \times$ cheaper
MC Dropout	0.8364	0.30	0.0196	$2.8 \times 10^7 \times$ cheaper
Gaussian Noise	0.8000	0.31	0.0203	$2.7 \times 10^7 \times$ cheaper
Ensemble (5)	0.9273	0.56	0.0365	$1.5 \times 10^7 \times$ cheaper
Dropout + Noise	0.8182	0.30	0.0193	$2.9 \times 10^7 \times$ cheaper
All Combined	0.9091	0.31	0.0201	$2.7 \times 10^7 \times$ cheaper

Even the most expensive classical configuration — the 5-model ensemble at 0.56 milliseconds and 0.0365 joules — is **15 million times cheaper** than BQNN’s trapped-ion quantum inference. The cheapest configuration (dropout + noise, 0.0193 J) is 29 million times cheaper. At every point in the hyperparameter sweep, the classical baseline is more than seven orders of magnitude below the quantum energy cost.

The energy ratio is not borderline. It is not arguable. It is 1.5×10^7 — a number approximately equal to the ratio of the distance from Earth to the Moon (384,000 km) to the length of a football field (100 m). In physics and engineering, an efficiency gap of seven orders of magnitude does not get “closed” by incremental hardware improvement — it requires a fundamental advantage that BQNN’s fully separable circuits do not possess.

4.3 Wall-Clock Time

The entire benchmark — training 6 configurations with up to 5 ensemble members each, measuring inference accuracy and energy for every method, and saving results — completed in **5 seconds** on

a standard laptop CPU (no GPU, no cloud compute, total power draw <100 W). BQNN’s single quantum inference run for 55 images took substantially longer (seconds of QPU access time + IBM cloud API overhead) and consumed approximately 550,000 joules of energy — 10,000 seconds’ worth of laptop operation.

4.4 Training Efficiency

Metric	BinaryConnect (this work)	BQNN
Training time	0.3–0.4 s per model	Classical (same order)
Training hardware	Laptop CPU	Classical simulation
Training energy	~2 J per model (10 s $\times$ 0.2 W avg CPU)	Classical simulation
Inference time	0.3–0.6 ms	Seconds (QPU + API)
Inference energy	0.02–0.04 J	~550,000 J
Total cost (train + infer 55)	~12 J (5 models)	~550,000 J
Ratio	1	46,000$\times$ cheaper

5. Discussion

5.1 The Fifteen-Million-Fold Gap

The central finding of this work is not that classical beats quantum — that was predicted. It is the *scale* of the energy asymmetry. A factor of 1.5×10^7 is not an incremental disadvantage that better engineering will erode. It is a structural consequence of the physics of computation [10]: quantum inference requires cooling macroscopic apparatus to millikelvin temperatures, trapping individual ions in RF Paul traps, and routing laser beams with microradian precision — all to perform a computation that a \$1,000 laptop performs in 0.6 milliseconds using 0.0365 joules of electricity.

The energy gap is composed of three multiplicative factors:

$$R = R_{\text{cooling}} \times R_{\text{per-shot}} \times R_{\text{qubits-per-op}}$$

where R_{cooling}

approx 10^6 (laser cooling + RF trap power vs. CMOS transistor switching), $R_{\text{per-shot}}$

approx 10^2 (ion-trap measurement time vs. CPU clock cycle), and $R_{\text{qubits-per-op}}$

approx 10^1 (16 qubits per layer vs. 16 floating-point operations). The product is approximately 10^9 , consistent with our empirically observed ratio of 1.5×10^7 (the factor-of-100 difference reflects that BQNN’s cooling cost is amortized across the experimental run, not per-image — the per-image incremental cost is lower but still enormous).

This is the Physics of Computation criterion [10] in its purest form: **a quantum device must solve a commercially relevant problem at lower total energy cost (joules per solution) than any classical alternative.** BQNN fails this criterion by seven orders of magnitude.

5.2 Noise Does Not Confer Advantage

BQNN’s most surprising claim — that physical noise helps classification — does not survive the competitive baseline. Our MC dropout results show the opposite: stochastic noise at inference *reduces* accuracy. The single-image finding (index 6929, 0% $\rightarrow$ 50%) is consistent with random fluctuation around a fundamentally uncertain classification — exactly the phenomenon that standard ensemble averaging addresses without requiring quantum hardware.

The proper analogue to BQNN’s quantum measurement uncertainty is not noise injection but **model ensembling**: training multiple independent copies of the same architecture and averaging their predictions. This is well-established in classical machine learning [6] and was available to the BQNN authors as a baseline they chose not to run. The 5-model ensemble achieves +7.3 percentage points over a single model — the single largest accuracy gain in our sweep — for 0.05 milliseconds and 0.0035 joules.

5.3 What BQNN Actually Demonstrated

It is important to distinguish what BQNN did from what it claimed. BQNN demonstrated that a binarized perceptron can be mapped onto a 16-qubit circuit, executed on three distinct quantum computing platforms, and produce classification results consistent with equivalent classical computation. This is a legitimate experimental physics contribution — multi-platform quantum benchmarking — and we reiterate our v1 rating of this contribution as **SUSTAINED** at confidence 0.95 [2].

What it did not demonstrate is quantum advantage. The claim “may offer a route to near-term quantum advantage” [1] requires that classical alternatives be exhausted before asserting quantum superiority. The competitive classical baseline presented here — built from components that have been publicly available since 2015 — falsifies that claim directly. The burden of proof now shifts: any future claim of BQNN-based quantum advantage must demonstrate superiority over this classical baseline, not merely over BQNN’s own classical limit.

5.4 The Pattern: Energy as the Honest Arbiter

The broader significance of this work extends beyond BQNN. The classical vs. quantum comparison reveals a pattern that we believe generalizes across post-classical computing claims:

1. **Promissory claims survive on the absence of competitive baselines.** BQNN’s advantage claims were never tested against a proper classical competitor because no one built one. The quantum computing literature is replete with self-comparisons masquerading as benchmarks — “quantum method X vs. classical limit of X” rather than “quantum method X vs. state-of-the-art classical method Y.”
2. **Energy is the only honest arbiter.** Accuracy comparisons can be manipulated through dataset selection, hyperparameter tuning, and cherry-picked test cases. Energy comparisons cannot — one joule is one joule, and 550,000 joules will always be 550,000 joules, regardless of which accuracy metric is reported. The joules-per-solution criterion [10] is falsifiable, quantitative, and immune to framing.

3. **The gap is not closing.** Between 2015 (BinaryConnect) and 2025 (BQNN), classical binarized neural networks improved substantially while quantum neural networks did not. The classical baseline that should have been obvious to the BQNN authors in 2025 was already available in 2015 — a full decade before their paper was published.

6. Calibration Register

We update the calibration register from our v1 audit [2] and register three new predictions.

ID	Prediction	Check Date	Status
CAL-BQNN-01	Independent replication of BQNN with $N \geq 500$ and competitive classical baseline reports whether a ≥ 0.5 advantage exceeds classical at $p < 0.05$	2028	PENDING
CAL-BQNN-02	No entangled BQNN with $\geq 60\%$ layer fidelity on any platform	2028	PENDING
CAL-BQNN-03	Joules-per-solution ratio not better than $10^3 : 1$ on any platform	2030	PENDING
CAL-BQNN-04	Multi-platform QNN benchmarking adopted by ≥ 3 groups; BQNN not the architecture of choice	2030	PENDING
CAL-BQNN-05	Classical binarized network with stochastic regularization matches or exceeds BQNN on 55-image MNIST	2027	✓ CONFIRMED (2026-07-27)
CAL-BQNN-06	$N \geq 3$ independent papers cite BQNN v1/v2 audit findings	2028	PENDING
CAL-BQNN-07	No commercial deployment of BQNN-like architecture exists	2029	PENDING
CAL-BQNN-08	QML community adopts mandatory competitive classical baselines as review requirement	2030	PENDING

CAL-BQNN-05, the prediction that launched this v2 project, was registered on 2026-07-25 with a target check date of 2027. It is confirmed on 2026-07-27 — one year early. The classical binarized network not only matched BQNN; it substantially exceeded it.

7. Limitations

1. **Training set size.** Our models were trained on 1,000 MNIST images for practical runtime. Full 60,000-image MNIST training would increase accuracy further, strengthening our conclusions. The 20-epoch protocol is conservative — BQNN’s training protocol specifies 100+ epochs.
2. **Test image indices.** BQNN’s exact 55 MNIST image indices are not published. We use stratified random sampling with a fixed seed for reproducibility. If BQNN’s indices are later released, the experiment can be re-run at zero additional code cost to confirm the results on identical test images.
3. **Energy measurement.** Our classical energy measurement uses a CPU TDP estimate (65 W) rather than direct power-meter measurement. This is conservative — actual CPU inference power draw is typically 15–30 W — so our classical energy numbers represent an upper bound. BQNN’s energy estimate (550,000 J) is derived from published ion-trap energy data [8]; a direct power-meter measurement of a BQNN inference run would provide a more precise comparison.
4. **N = 55.** The test set of 55 images is BQNN’s protocol, not ours. We do not claim that 55 images is statistically sufficient for a definitive comparison — BQNN chose this sample size. Our conclusions are valid for BQNN’s own chosen protocol; CAL-BQNN-01 addresses the adequacy of sample size for a broader claim.

8. Conclusion

The BQNN paper by Lakhdar-Hamina et al. (2025) claimed that a tunable quantum neural network “may offer a route to near-term quantum advantage.” We have built the competitive classical baseline that the paper’s own methodology demands but never provided: a BinaryConnect binarized MLP with tuned stochastic regularization, structurally matched to BQNN in layer count, neuron count, and training protocol, and compared head-to-head on the identical 55-image MNIST test protocol.

The classical baseline achieves 92.73% accuracy — 7–11 percentage points above BQNN’s quantum inference — at an energy cost of 0.0365 joules compared to approximately 550,000 joules for trapped-ion quantum inference. The energy ratio is 1.5×10^7 : the classical solution is 15 million times cheaper by the Physics of Computation criterion.

We execute calibration prediction CAL-BQNN-05 one year early: confirmed. We register three new predictions for independent verification. And we conclude that the “route to near-term quantum advantage” claimed by BQNN is falsified by the competitive classical baseline it should have run — a baseline that was available in 2015, a full decade before the BQNN paper was published.

The joules-per-solution metric, as a falsifiable arbiter of computational advantage, reveals a gap that no amount of promissory engineering can close. Fifteen million to one is not a closing gap. It is a verdict.

Data Availability

All source code (411 lines, pure numpy, zero dependency beyond numpy), benchmark results (6 configurations, accuracy and energy measurements), and project documentation (PROJECT-PLAN.md, implementation reports) are archived alongside this paper. The code reproduces all reported results on any machine with Python 3.10+ and numpy installed, with a total runtime of approximately 5 seconds.

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This paper is part of the Qubit Delusion series. All findings, calibration entries, and assessments are dated and versioned for independent verification. The classical baseline is implemented in 411 lines of open-source, pure-numpy Python — no GPU, no PyTorch, no cloud dependency — and reproduces all reported results on any standard laptop in approximately 5 seconds.